

Phylogenetics - IV

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4.1.2 Neighbor Joining (NJ) Method

~~UPGMA~~

- ~~Assumes a 'molecular clock'~~
 - ~~Rate of change is constant in sister species~~
 - ~~Hence branch lengths are equal~~
- Starts from a distance matrix
- Pairs up the most closely related taxa
- Treat this pair as new taxa

NJ method

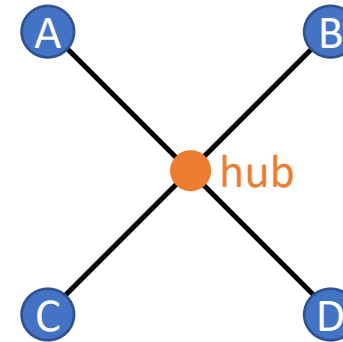
Therefore, taxa is not equidistant from the root



NJ corrects for unequal evolutionary rates between sequences using a conversion step.



	A	B	C	D
A	-	17	21	27
B	-	-	12	18
C	-	-	-	14
D	-	-	-	-



Round 1 Step 1

Compute r'_i for each terminal node

	A	B	C	D
A	-	17	21	27
B	-	-	12	18
C	-	-	-	14
D	-	-	-	-

	A	B	C	D	r'_i
A	-	17	21	27	32.5
B	-	-	12	18	23.5
C	-	-	-	14	23.5
D	-	-	-	-	29.5

$$r'_i = \frac{\sum_j D_{i,j}}{n-2}$$

no. of taxa
 $4-2=2$
A
B
C
D

$$r'_A = (17 + 21 + 27) / 2 = 32.5$$
$$r'_B = (17 + 12 + 18) / 2 = 23.5$$
$$r'_C = (21 + 12 + 14) / 2 = 23.5$$

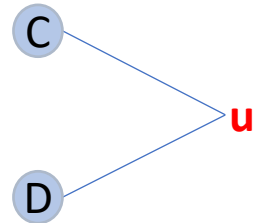


Round 1 Step 2

Compute $d'_{i,j}$ for each terminal node

d	A	B	C	D	r'_i
A	-	17	21	27	32.5
B	-	-	12	18	23.5
C	-	-	-	14	23.5
D	-	-	-	-	29.5

d'	A	B	C	D
A	-	-39	-35	-35
B	-	-	-35	-35
C	-	-	-	-39
D	-	-	-	-



$$\underline{d'_{i,j} = d_{i,j} - r'_i - r'_j}$$

$$\begin{aligned} d'_{ab} &= 17 - 32.5 - 23.5 = -39 \\ d'_{ac} &= 21 - 32.5 - 23.5 = -35 \end{aligned}$$

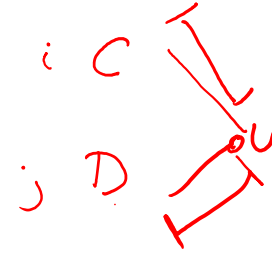
$$d'_{ad} = 27 - 32.5 - 29.5 = -35$$

$$d'_{bc} ; d'_{bd} ; d'_{cd}$$

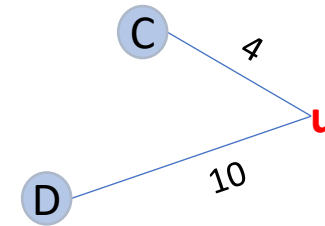


Round 1 Step 3

Calculate branch lengths



d	A	B	C	D	r'_i
A	-	17	21	27	32.5
B	-	-	12	18	23.5
C	-	-	-	14	23.5
D	-	-	-	-	29.5



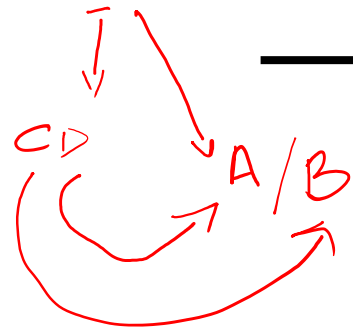
$$\checkmark V_i = 0.5 (d_{i,j}) + 0.5 (r'_i - r'_j) = 0.5 (14) + 0.5 (\underline{23.5} - \underline{29.5}) = 4 \checkmark$$
$$\checkmark V_j = 0.5 (d_{i,j}) + 0.5 (r'_j - r'_i) = 0.5 (14) + 0.5 (\underline{29.5} - \underline{23.5}) = 10 \checkmark$$



Round 2

	A	B	C	D
A	-	17	21	27
B	-	-	12	18
C	-	-	-	14
D	-	-	-	-

$$d'_{ij,k} = (d_{i,k} + d_{j,k} - d_{i,j})/2$$

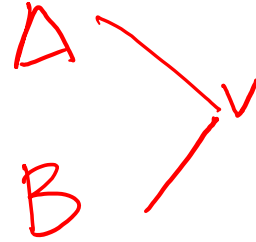


	A	B	CD	r'_i
A	-	17	17	34
B	-	-	8	25
CD	-	-	-	25

$$d'_{CD,A} = (21 + 27 - 14)/2 = 17$$

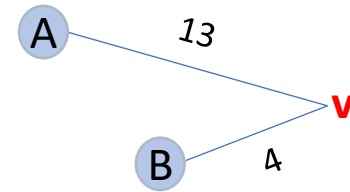
$$d'_{CD,B} = (12 + 18 - 14)/2 = 8$$





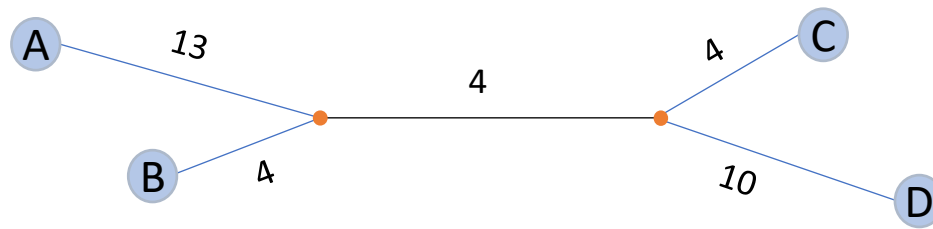
	A	B	CD	r_i
A	-	17	17	34
B	-	-	8	25
CD	-	-	-	25

	A	B	CD
A	-	-42	-42
B	-	-	-42
CD	-	-	-



Round 3

	AB	CD
AB	-	4
CD	-	-



	A	B	C	D
A	-	17	21	27
B	-	-	12	18
C	-	-	-	14
D	-	-	-	-

